1 Limits of inverse functions

Recall that a function defined on an interval is said to be *monotonic* if it is either increasing or decreasing.

Theorem 1.1 (Continuity of inverse functions)**.** *If f is a strictly monotonic function on an interval I, then $f^{-1} : f(I) \to I$ is continuous.*

Proof. It is sufficient to prove this for f strictly increasing, for, if f is strictly decreasing, then $-f$ is strictly increasing and then continuity of $-f$ implies continuity of f, as $-f$ is a constant multiple of a continuous function. Suppose now that f is strictly increasing. To prove that $f^{-1} : f(I) \to I$ is continuous on $f(I)$, we need to show that, for any $b \in f(I)$, $\lim_{y\to b} f^{-1}(y) = f^{-1}(b)$. That is, we need to show that given any $\varepsilon > 0$ there exists $\delta > 0$ such that $|y-b| < \delta$ implies $|f^{-1}(y) - f^{-1}(b)| < \varepsilon$. Given $b \in f(I)$ and $\varepsilon > 0$, suppose first that $a = f^{-1}(b)$ is an interior point of I. Then there exist $c, d \in I$ with

$$a - \varepsilon < c < a < d < a + \varepsilon. \tag{1.2}$$

Since f is strictly increasing, it follows that

$$f(c) < f(a) = b < f(d),$$

so we have $b - f(c) > 0$ and $f(d) - b > 0$. Choose $\delta = \min\{b - f(c), f(d) - b\}$. Then $|y-b| < \delta$ implies, in particular, that $y - b > f(c) - b$ and therefore $y > f(c)$, and $y - b < f(d) - b$ and therefore $y < f(d)$, so we have

$$f(c) < y < f(d),$$

and therefore

$$c < f^{-1}(y) < d$$

since f is strictly increasing. By (1.2),

$$a - \varepsilon < c < f^{-1}(y) < d < a + \varepsilon,$$

and therefore $|f^{-1}(y) - a| = |f^{-1}(y) - f^{-1}(b)| < \varepsilon$. This proves that f^{-1} is continuous when $f^{-1}(b)$ is an interior point of I. The proof when $f^{-1}(b)$ is an endpoint of I is similar. □

Corollary 1.3 (Continuity of logarithmic functions)**.** *For any $b \in (0,1) \cup (1,\infty)$, $\log_b(x)$ is continuous on its domain.*

Proof. The function $f(x) = b^x$ is strictly monotonic on $(-\infty, \infty)$ (strictly increasing when $b > 1$ and strictly decreasing when $0 < b < 1$), so by Theorem 1.1 $\log_b(x)$ is continuous on the image of e^x, which is $(-\infty, \infty)$. □

There are two other limits of interest of logarithmic functions.

Proposition 1.4. *If $b > 1$, then*

$$\lim_{x\to\infty} \log_b(x) = \infty,$$
$$\lim_{x\to 0^-} \log_b(x) = -\infty.$$

If $0 < b < 1$, then

$$\lim_{x\to\infty} \log_b(x) = -\infty,$$
$$\lim_{x\to 0^-} \log_b(x) = \infty.$$

Proof. Suppose $b > 1$. Given $\varepsilon > 0$, let $\delta = b^{\varepsilon} > 0$. Then, since $\log_b(x)$ is strictly increasing, $x > b^{\varepsilon}$ implies $\log_b(x) > \log_b(b^{\varepsilon}) = \varepsilon$. This proves that $\lim_{x\to\infty} \log_b(x) = \infty$.

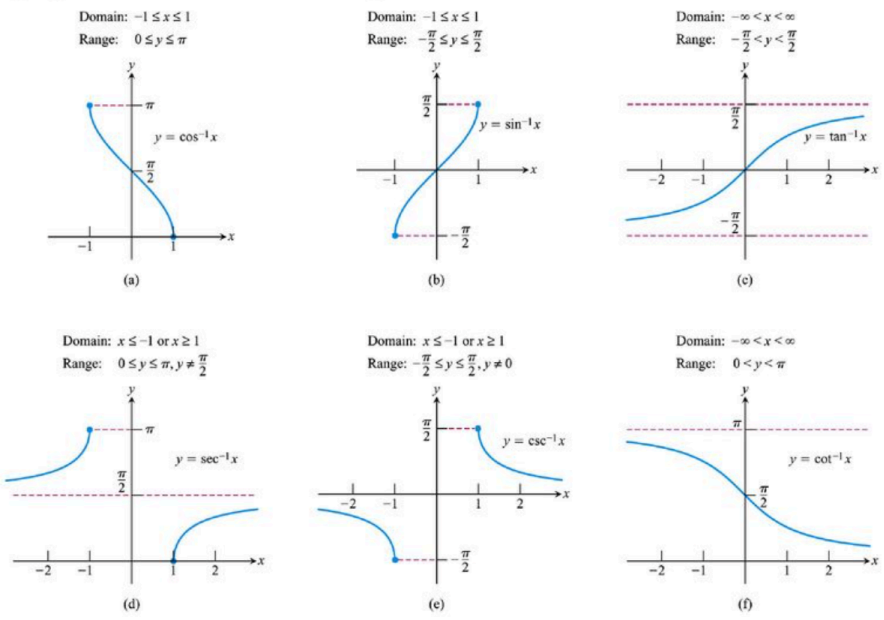
Given $\varepsilon > 0$, now take $\delta = b^{-\varepsilon} > 0$. Then since $\log_b(x)$ is strictly increasing, $0 < x < \delta$ implies $\log_b(x) < \log_b(b^{-\varepsilon}) = -\varepsilon$.

The proof in the case of $0 < b < 1$ is similar, using the fact that $\log_b(x)$ is strictly decreasing. □

Corollary 1.5 (Continuity of inverse trig functions)**.** *The inverse trig functions are all continuous on their domain.*

Proof. Each of the trig functions is strictly monotonic on the appropriate restricted domain (e.g., $\sin(x)$ is strictly increasing on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, etc.), so by Theorem 1.1 the inverse of each of these functions is continuous on their domain. □

Proving the various limits at infinity and infinite limits of inverse trig functions is straightforward but tedious. It is best to just memorize the graphs.



2 Composite Function Theorem

Theorem 2.1 (Composite Function Theorem)**.** *If f is continuous at L and $\lim_{x\to a} g(x) = L$, then*

$$\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right) = f(L).$$

Proof. To prove that $\lim_{x\to a} f(g(x)) = f(L)$, we need to show that, given $\varepsilon > 0$, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(g(x)) - f(L)| < \varepsilon$. Since f is continuous at L, corresponding to ε there exists $\delta_1 > 0$ such that $0 < |g(x) - L| < \delta_1$ implies $|f(g(x)) - f(L)| < \varepsilon$. Since $\lim_{x\to a} g(x) = L$, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|g(x) - L| < \delta_1$. Thus, $0 < |x-a| < \delta$ implies $|g(x) - L| < \delta_1$ implies $|f(g(x)) - f(L)| < \varepsilon$, which proves that $\lim_{x\to a} f(g(x)) = f(L)$. □

Corollary 2.2 (A composition of continuous functions is continuous)**.** *If g is continuous at a and f is continuous at $g(a)$, then $f \circ g$ is continuous at a.*

Proof. By the Composite Function Theorem

$$\begin{aligned}
\lim_{x\to a}(f\circ g)(x) &= \lim_{x\to a} f(g(x)) && \text{(by definition of } (f\circ g)(x)) \\
&= f(\lim_{x\to a} g(x)) && \text{(by the Composite Function Theorem)} \\
&= f(g(a)) && \text{(since } g \text{ is continuous at } a) \\
&= (f\circ g)(a) && \text{(by definition of } (f\circ g)(x)).
\end{aligned}$$

This proves that $f \circ g$ is continuous at a. □

3 Exercises

1. Where is the function $f(x) = \dfrac{\ln(x) + \tan^{-1}(x)}{x^2 - 1}$ continuous?

2. Evaluate $\lim_{x\to 1} \arcsin\left(\dfrac{1-\sqrt{x}}{1-x}\right)$.

3. Where is the function $h(x) = \sin(x^2)$ continuous?

4. Where is the function $F(x) = \ln(1+\cos(x))$ continuous?

5. Evaluate $\lim_{x\to\pi} \sin(x+\sin(x))$.

6. Evaluate $\lim_{x\to 2} \arctan\left(\frac{x^2-4}{3x^2-6x}\right)$